Detecting Clickbait and Sensationalism in Ghanaian Online News Headlines

Mackey Kumi (22025755)

DCIT 316: Computational Models for Social Media Mining

Department of Computer Science, University of Ghana

October 2026

Code, data and notebooks: https://github.com/Mackey-Kumi/ghana-clickbait-detection

Abstract

Most clickbait detection research has been done on English-language news from the US and UK, and I could not find a labelled dataset of Ghanaian headlines to work with. For this project I built one. I scraped 1,000 headlines from seven Ghanaian news sites (four mainstream outlets and three entertainment/gossip sites), labelled every one of them as Clickbait, Informative or Ambiguous using a written codebook, and then compared three ways of detecting clickbait automatically: a TF-IDF + Logistic Regression baseline, a fine-tuned DistilBERT model, and a zero-shot classifier (`facebook/bart-large-mnli`) that gets no training at all. Fine-tuned DistilBERT did best, with a macro F1 of 0.714 and Clickbait recall of 0.744, against 0.647 and 0.590 for the baseline. It also held up better when tested on a whole outlet it had never seen during training. The zero-shot model was very sensitive to how I worded its labels: with the plain labels "clickbait" and "informative" it did not catch a single clickbait headline, but with longer descriptive labels it came close to the baseline. A second annotator who labelled 250 headlines without seeing the codebook agreed with me only slightly above chance (Cohen's $\kappa = 0.057$), which I think says more about how vague "clickbait" is as an everyday word than about the labels themselves. All code, data and results are in the GitHub repository linked above.

1. Introduction

Anyone who reads Ghanaian news online has seen headlines like "Popular Actress Breaks Down In Tears After What Her Husband Did" or "See What Sarkodie Said About Shatta Wale That Has Got Everyone Talking." These headlines are written to make you click, not to tell you what happened. They work by holding back the key piece of information (who, what, or why), so the only way to find out is to open the article and load the ads on the page.

Clickbait is not just annoying. When a large share of the headlines people scroll past on Facebook, WhatsApp and X are built this way, it gets harder to tell real news from noise, and sensational framing can make ordinary stories look like scandals. In Ghana a lot of news is consumed through shared links on social media, where the headline is often the only part people actually read, so headlines carry a lot of weight.

There has been a good amount of research on detecting clickbait automatically (Section 2), but almost all of it uses English headlines from Western outlets like BuzzFeed, The Guardian or the New York Times. Ghanaian clickbait has its own style. It leans on vague celebrity references ("popular Ghanaian actress", "well-known pastor"), quotes taken out of context, and

phrases like "breaks down in tears" or "puts his huge house on display." I wanted to know whether the methods that work on Western data would work here too, and I also wanted to find out what it actually takes to build a labelled dataset for this kind of problem from scratch.

My research questions were:

1. **RQ1:** How well can a simple, interpretable model (TF-IDF + Logistic Regression) separate clickbait from informative Ghanaian headlines?
2. **RQ2:** Does fine-tuning a small transformer model (DistilBERT) do noticeably better?
3. **RQ3:** Can a zero-shot model, with no training on my data at all, do the job?
4. **RQ4:** Do these models learn clickbait *style*, or do they just learn which outlet a headline came from?

My contributions are (a) a new dataset of 1,000 labelled Ghanaian headlines, (b) an annotation codebook with Ghana-specific clickbait signals that I refined three times while labelling, and (c) a comparison of the three modelling approaches on the same test set, including a held-out-outlet test that addresses RQ4.

2. Related Work

Clickbait detection. Chen, Conroy and Rubin (2015) described clickbait as a form of "false news" and suggested combining lexical, semantic and image cues to detect it. Potthast, Köpsel, Stein and Hagen (2016) built one of the first clickbait corpora from Twitter posts by news accounts and trained a classifier on hand-crafted features such as sentence structure, word n-grams and specific "bait" phrases. Chakraborty, Paranjape, Kakarla and Ganguly (2016) collected around 15,000 headlines from sites like BuzzFeed and Upworthy (clickbait) and Wikinews and the New York Times (non-clickbait), and reached about 93% accuracy with an SVM. They also built a browser extension that warned users about clickbait. The Clickbait Challenge 2017 (Potthast, Gollub, Hagen & Stein, 2018) turned the task into a shared competition with around 38,000 crowd-annotated tweets, and many of the strongest systems used neural networks.

One thing I noticed in these papers is that the clickbait and non-clickbait examples often come from *different sources*. If all your clickbait comes from BuzzFeed and all your non-clickbait comes from the New York Times, a classifier can score very well just by learning which publication wrote the headline. That is one reason I decided to label every headline individually instead of labelling by outlet, and why I added the held-out-outlet test in Section 5.4.

Why clickbait works. Most of this work builds on Loewenstein's (1994) "information gap" theory of curiosity: when we notice a gap between what we know and what we want to know, we feel a pull to close it. Clickbait headlines create that gap on purpose. My codebook (Section 3.3) is basically a list of the specific ways Ghanaian headlines create these gaps.

Transformers and zero-shot classification. BERT (Devlin, Chang, Lee & Toutanova, 2019) made it standard to take a large pre-trained language model and fine-tune it on a small labelled dataset. DistilBERT (Sanh, Debut, Chaumond & Wolf, 2019) is a smaller version of BERT that keeps most of its accuracy while being about 40% smaller and faster, which made it realistic for a student project without a GPU. For zero-shot classification, Yin, Hay and Roth (2019) showed that a model trained on natural language inference (deciding whether one sentence implies another) can classify text into labels it was never trained on. You turn each label into a sentence like "This text is about sports." and check whether the model thinks the input implies it. The `facebook/bart-large-mnli` model I used is BART (Lewis et al., 2020) fine-tuned on the MultiNLI dataset (Williams, Nangia & Bowman, 2018) and is one of the most widely used models for this.

AI-assisted annotation. Gilardi, Alizadeh and Kubli (2023) found that ChatGPT could match or beat crowd workers on several text annotation tasks. That made me think an AI model could speed up my own labelling, but I did not want to just trust it, so I used it only to produce draft labels that I then checked one by one (Section 3.3). As it turned out, the AI drafts missed most of the clickbait, which supports being careful here.

I did not find any published clickbait dataset or study focused on Ghanaian news, which is the gap this project tries to fill.

3. Data

3.1 Choosing the sources

I wanted a mix of outlets that I expected to be mostly informative and outlets that I expected to have a lot of clickbait. I ended up with seven:

Outlet	Type I expected	Headlines collected
Myjoyonline	Mainstream / credible	155
Graphic Online	Mainstream / credible	35
Citinewsroom	Mainstream / credible	155
Ghana News Agency (GNA)	Mainstream / credible	155

Outlet	Type I expected	Headlines collected
GhPage	Entertainment / gossip	167
ZionFelix	Entertainment / gossip	167
Yen.com.gh	Entertainment / gossip	166
Total		**1,000**

Table 1. Sources and number of headlines collected from each (500 per group).

Getting to this list took more work than I expected. My original plan had other sites on it. Two popular gossip sites, ghgossip.com and ameyawdebrah.com, block AI crawlers in their `robots.txt` files, so I left them out instead of trying to get around the block. I also considered adomonline.com, but it belongs to the same media group as Myjoyonline and its headlines read more like mainstream news, so including it in the gossip group would have been misleading.

Graphic Online only gave me 35 headlines because its category pages don't actually paginate (page 2 shows the same stories as page 1), and only two of its sections used a consistent page layout I could scrape. I took more headlines from the other three mainstream outlets to keep the two groups at 500 each.

3.2 Scraping

I wrote the scraper in Python using `requests` and `BeautifulSoup`. For six of the sites I scraped the category/listing pages. Yen.com.gh doesn't render its listing pages on the server, so for Yen I read its XML sitemap, filtered it to celebrity/entertainment articles, and fetched each article page to get the headline. The scraper waits 1.5 seconds between requests, identifies itself with a proper User-Agent string, retries failed requests a couple of times, and removes duplicate URLs. I only collected the headline text and the article URL, no article bodies and no personal data. Scraping was done on 25 August 2026. Before running the full scrape I wrote unit tests for the parsing and crawling logic (they are in `tests/`).

3.3 Labelling

Codebook. I wrote a codebook (`docs/ANNOTATION_CODEBOOK.md`) with three labels:

- **Clickbait:** the headline uses curiosity gaps, emotional manipulation, vague references, excessive capitals or false urgency to get clicks instead of informing.
- **Informative:** the headline tells you what the story actually is, even if the story itself is dramatic.
- **Ambiguous:** genuinely can't decide. These were excluded from training.

The codebook also lists Ghana-specific signals such as vague celebrity references ("popular Ghanaian actress"), emotional bait ("breaks down in tears"), and curiosity-gap phrases ("see what she said").

AI pre-labelling plus my own check. Labelling 1,000 headlines alone takes a long time, so I had an AI model (Claude) give every headline a draft label and a one-line reason based on the codebook. I then went through all 1,000 myself with a small command-line tool I wrote (`annotate/verify.py`) that shows the headline, the draft label and the reason, and lets me accept or change it. This took me about five hours. My labels are the final ground truth, not the AI's.

I changed the codebook three times while labelling. As I went through the drafts I kept disagreeing with the AI in the same ways, and each time I realised the codebook itself wasn't clear enough. I made three changes:

1. **Disturbing topics are not automatically clickbait, and they are not automatically informative either.** A headline about something terrible can be perfectly informative if it says who did what. The test is whether the headline gives you the substance, not how upsetting it is.
2. **A quote is not automatically informative.** The AI kept marking quote headlines as informative because the quote is real and complete. But a headline like "My Manager Wanted To Rape Me – Ruby Of Date Rush" tells you nothing about what actually happened: no report, no named person accused, no outcome. You still have to click. Compare that with "'I apologise' – Dr Yamson admits MTN Ghana was late to condemn xenophobic attacks", where the framing around the quote adds a real fact (a named executive and company, apologising for a specific failure).
3. **Vague hype words are not description.** "Kofi Kinaata Puts His Huge House And Plush Living Room On Display (Watch Video)" names a real person and is a normal sentence, but "huge" and "plush" can't be checked. The only point of the headline is to get you to watch the video. Compare "Davido Spoils Bride Chioma with $300K Richard Mille Diamond Watch at Miami Wedding": also a show-off story, but you learn real, checkable facts without clicking.

How often I overrode the AI. I changed 167 of the 1,000 draft labels (16.7%). The table below shows the direction:

AI draft → my label	Ambiguous	Clickbait	Informative	Total
Ambiguous	10	19	1	30
Clickbait	0	32	2	34
Informative	0	**145**	791	936
Total	10	196	794	1,000

Table 2. AI draft labels (rows) against my final labels (columns).

The big number is the 145 headlines the AI called Informative that I called Clickbait. The AI only flagged 32 of the 196 headlines I ended up labelling Clickbait, so on its own it would have missed most of the clickbait in the dataset. Nearly all of the misses fall into the three gaps above. This is a big reason I would not trust AI labels for this task without a human going through them.

Final label distribution.

Outlet	Informative	Clickbait	Ambiguous
Citinewsroom	148	6	1
GNA	153	2	0
Graphic Online	33	2	0
Myjoyonline	146	5	4
GhPage	89	75	3
ZionFelix	98	67	2
Yen.com.gh	127	39	0
Total	**794 (79.4%)**	**196 (19.6%)**	**10 (1.0%)**

Table 3. Final labels by outlet.

Two things stand out. First, the dataset is imbalanced, roughly 80% Informative to 20% Clickbait, which matters a lot for evaluation (Section 4.4). Second, even the gossip sites are mostly *not* clickbait under my codebook: only 181 of their 500 headlines (36%) were Clickbait. So "this site is a gossip site" and "this headline is clickbait" are very different things, which is why I labelled headline by headline.

3.4 Second annotator and inter-annotator agreement

To check how reliable the labels were, I gave a second person 250 headlines picked at random (stratified by outlet) in an Excel sheet with a dropdown for the label. They only saw an ID and the headline text. They didn't see the outlet, the URL, the AI's draft or my labels.

The result was **Cohen's κ = 0.057** (Cohen, 1960), with raw agreement of 54.6% (136 of 249 paired items; one row was left blank). On Landis and Koch's (1977) scale this is only "slight" agreement, barely above chance.

I looked into why, and I don't think it means my labels are random. The second annotator labelled from the spreadsheet alone, without reading the codebook or being briefed on the criteria. Their disagreement with me was one-sided: they called 41.8% of headlines Clickbait while I called 16.5% of the same headlines Clickbait, and 86 of the 205 headlines I labelled Informative they labelled Clickbait. So they were using a much broader, everyday meaning of "clickbait", something closer to "any gossip or celebrity story", while I was applying the narrower definition in the codebook. Cohen's κ is normally used to check whether a *shared, written* labelling scheme can be applied consistently by different people. What I actually measured here is the gap between someone using the scheme and someone using their own intuition. I discuss what this means in Section 6.

4. Methods

4.1 Common setup

All three models use exactly the same data and the same test set, so their results can be compared fairly. I dropped the 10 Ambiguous headlines, leaving 990 (794 Informative, 196 Clickbait). I split these 80/20 into training (792) and test (198) sets, stratified by label, with a fixed random seed of 42. The loading and splitting code lives in one small module (`models/data.py`) that every notebook imports, so the split cannot accidentally differ between experiments. That module has its own unit tests.

4.2 Baseline: TF-IDF + Logistic Regression

I turned each headline into a TF-IDF vector of single words and two-word phrases (unigrams and bigrams), keeping terms that appear in at least two headlines (a vocabulary of 1,902 terms), and trained a Logistic Regression classifier with scikit-learn (Pedregosa et al., 2011). I chose this as the baseline because it is fast and easy to interpret: you can look at the learned weights and see which words push a headline towards Clickbait.

My first version didn't use any class weighting. It got 82.3% accuracy, which looked good, but it only caught 10.3% of the clickbait. It had basically learned to say "Informative" for almost everything. I switched to `class_weight="balanced"`, which makes mistakes on the smaller Clickbait class count for more during training. That version is the baseline I report.

4.3 Fine-tuned DistilBERT

I fine-tuned `distilbert-base-uncased` (Sanh et al., 2019) using the Hugging Face Transformers library (Wolf et al., 2020). My laptop has no dedicated GPU, so I ran the training on a free Kaggle notebook with an NVIDIA T4 GPU, controlled from my laptop with the Kaggle command-line tool. Settings: 4 epochs, batch size 16, learning rate 2e-5, weight decay 0.01, maximum length 64 tokens (headlines are short). To deal with the class imbalance in the same way as the baseline, I changed the loss function to a class-weighted cross-entropy. The model trained in about 14 seconds on the T4, and all eight runs (the main one plus seven held-out-outlet runs) took about 104 seconds in total.

4.4 Zero-shot classification with BART-MNLI

For the third approach I used Hugging Face's zero-shot classification pipeline with `facebook/bart-large-mnli`, again on Kaggle. This model is never trained on my data. For each headline it scores how strongly the headline "implies" a sentence built from each candidate label, and picks the higher one. I should say that my proposal called this a "zero-shot LLM", but BART-MNLI is not a chat model like ChatGPT. It is a 400-million-parameter model trained for natural language inference, and it is the standard choice for zero-shot classification.

I tried two ways of wording the labels:

- **Literal labels:** `"clickbait"` and `"informative"`, with the template "This news headline is {}." This was exactly what my proposal said I would do.
- **Descriptive labels:** `"a sensationalized clickbait headline designed to provoke curiosity"` and `"a factual, informative news headline"`, with the template "This is {}."

I added the second version after the first one failed badly (Section 5.3), to see whether the failure came from the model itself or from my wording.

4.5 Evaluation

Because the data is about 80/20, accuracy on its own is misleading: a model that always says "Informative" would already get about 80%. So I report **macro F1** (the average of the F1 scores of both classes, treating them equally) as the main metric, along with **Clickbait precision** (of the headlines flagged as clickbait, how many really were) and **Clickbait recall** (of the real clickbait headlines, how many were caught).

Held-out-outlet test (RQ4). With a random split, headlines from the same outlet appear in both training and test sets, so a model could do well just by recognising each outlet's writing style or recurring names. To test this I retrained the baseline and DistilBERT seven times, each time leaving one whole outlet out of training and testing on that outlet only. If a model really learned what clickbait looks like, it should still find clickbait on a site it has never seen.

5. Results

5.1 Main comparison

Model	Accuracy	Macro F1	Clickbait Precision	Clickbait Recall
Baseline, unweighted (not used)	0.823	0.543	1.000	0.103
Baseline, class-balanced	0.737	0.647	0.390	0.590
Fine-tuned DistilBERT	**0.783**	**0.714**	**0.468**	**0.744**
Zero-shot, literal labels	0.798	0.444	0.000	0.000
Zero-shot, descriptive labels	0.788	0.651	0.457	0.410

Table 4. Results on the same 198-headline test set (159 Informative, 39 Clickbait).

Fine-tuned DistilBERT is the best model on macro F1 and on Clickbait precision and recall. It caught 29 of the 39 clickbait headlines in the test set, compared to 23 for the baseline and 16 for the best zero-shot setup. The highest *accuracy* in the table actually belongs to the zero-shot model with literal labels, a model that caught no clickbait at all, which shows why accuracy isn't a useful metric for this dataset.

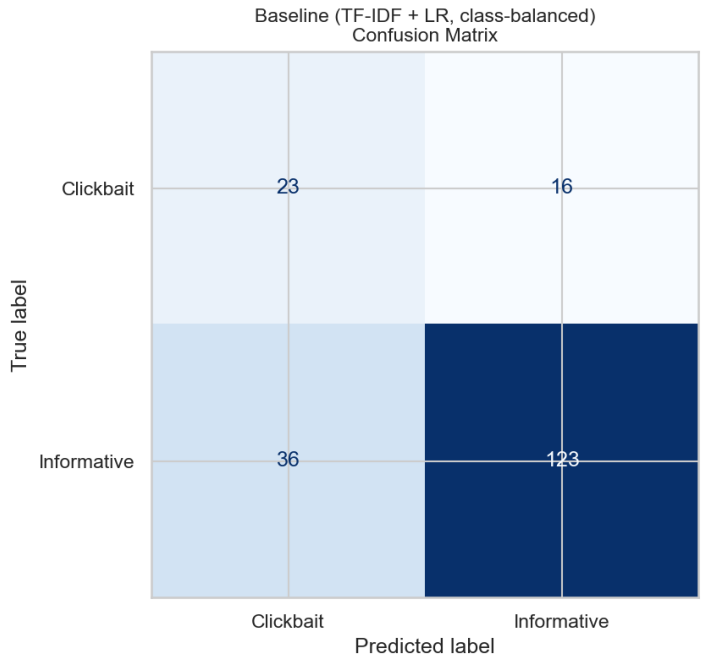


Figure 1. Baseline (TF-IDF + Logistic Regression, class-balanced) confusion matrix.

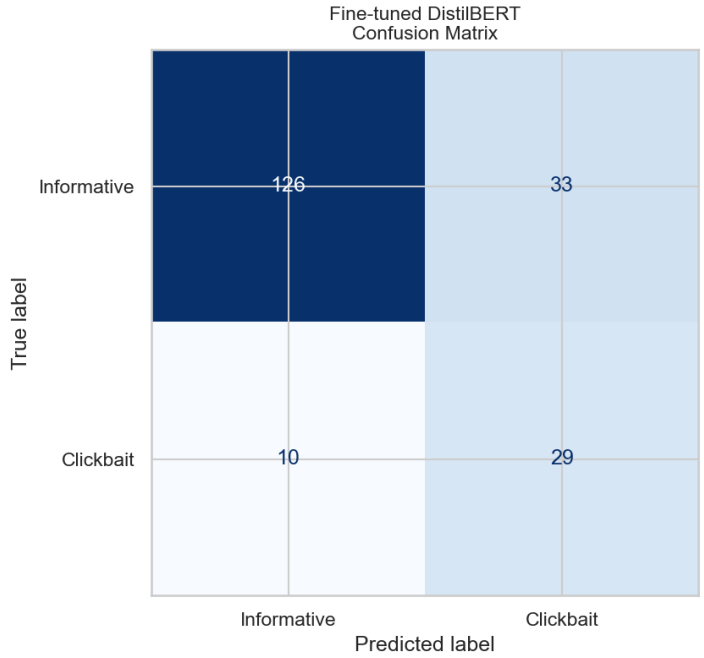


Figure 2. Fine-tuned DistilBERT confusion matrix: 126 Informative and 29 Clickbait correct, 33 false alarms, 10 missed clickbait.

DistilBERT's main weakness is false alarms: 33 informative headlines flagged as clickbait, which is why its Clickbait precision is only 0.468. In practice this means about half of what it flags is real clickbait. That's not good enough to block anything automatically, but it could be useful for something like a warning label.

5.2 What the baseline learned

Looking at the baseline's highest-weighted features was useful. Words pushing towards Clickbait included *video*, *photos*, *why*, *boyfriend*, *girlfriend*, *wife*, *lady*, *sparks*, *online*, *just*, *wealth* and pronouns like *his*, *her* and *she*. These fit the codebook well: "watch video" and "see photos" teasers, relationship gossip, and vague references to "a lady" or "a man" instead of a name. *Why* is a classic curiosity-gap word.

The Informative side is more worrying: *ghana*, *mahama*, *president mahama*, *npp*, *minister*, *national*, *2026*. These are words about the *topic* (politics, government, the current year), and those topics mostly appear on the mainstream outlets. So part of what the baseline learned is "politics means informative", which is a shortcut, not an understanding of clickbait. One feature, *abena* (a common Ghanaian first name), shows up on the Clickbait side, which is almost certainly because of a few specific celebrities who appear a lot on the gossip sites. This made me want to run the held-out-outlet test.

5.3 Zero-shot results and label wording

With the literal labels "clickbait" and "informative", the zero-shot model predicted "Informative" for 197 of the 198 test headlines and caught zero clickbait. When I ran it over the full dataset, its Clickbait recall was 0.000 for every one of the seven outlets, so this was not a quirk of the test split. The model simply never chose "clickbait".

With the descriptive labels, on the exact same test set, Clickbait recall went from 0.000 to 0.410 and macro F1 from 0.444 to 0.651, slightly ahead of the trained baseline (0.647). Nothing about the model changed, only the words in the candidate labels.

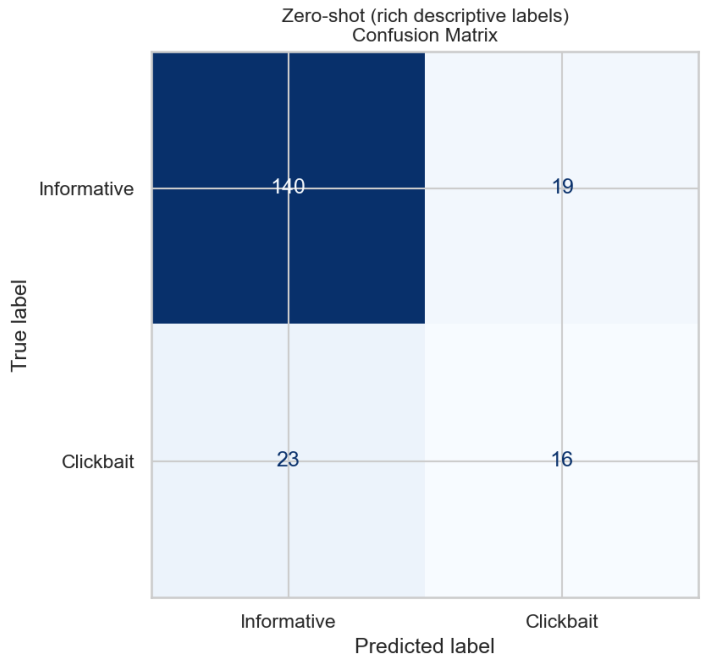


Figure 3. Zero-shot (BART-MNLI, descriptive labels) confusion matrix.

My guess for why the single word "clickbait" fails is that the NLI model was trained on ordinary sentence pairs and never really learned what "clickbait" means as a label. "This news headline is clickbait" might just not be something it can match to any headline, whereas the longer description uses plain words like "sensationalized" and "provoke curiosity" that it understands better.

5.4 Held-out-outlet test

Held-out outlet	Type	Baseline macro F1	DistilBERT macro F1	Baseline CB recall	DistilBERT CB recall
GhPage	Gossip	0.613	0.701	0.440	0.800
ZionFelix	Gossip	0.590	0.690	0.612	0.896
Yen.com.gh	Gossip	0.493	0.552	0.308	0.513
Gossip average		**0.565**	**0.648**	**0.453**	**0.736**
Citinewsroom	Mainstream	0.714	0.635	0.333	0.167
GNA	Mainstream	0.747	0.832	0.500	0.500
Graphic Online	Mainstream	0.826	0.826	0.500	0.500
Myjoyonline	Mainstream	0.579	0.635	0.400	0.600
Mainstream average		**0.716**	**0.732**	**0.433**	**0.442**

Table 5. Each row: model trained on the other six outlets, tested on this one.

The gossip outlets are the ones that matter here, since that's where almost all the clickbait is. When the baseline has never seen an outlet, its Clickbait recall drops from 0.590 (random split)

to 0.453 on average. That supports my worry from Section 5.2 that it was partly relying on outlet-specific words. DistilBERT holds up much better: its average recall on unseen gossip outlets is 0.736, almost the same as its 0.744 on the random split. It still finds about three-quarters of the clickbait on a site it has never trained on, which suggests it learned something closer to clickbait *style*.

Yen.com.gh was the hardest outlet for both models. I think this is because its headlines are written in a more polished, mainstream-looking style (it's a bigger, more established site) while still being mostly celebrity content, so its clickbait is subtler.

I would not read much into the mainstream rows. Those outlets only have 2 to 6 clickbait headlines each, so recall jumps in big steps (one headline out of six is 0.167), and the macro F1 numbers mostly reflect how few false alarms the model raises.

6. Error Analysis

To see *what kind* of headlines the models get wrong, I went through every mistake the baseline made on the test set. I used the baseline because its predictions can be regenerated locally from the notebook. For DistilBERT, I only saved the summary metrics from Kaggle, not per-headline predictions (see Limitations), but its confusion matrix (Figure 2) has the same shape: far more false alarms (33) than missed clickbait (10).

False alarms (36 headlines, Informative wrongly flagged as Clickbait). 33 of the 36 came from the three gossip sites (Yen 13, ZionFelix 11, GhPage 9). They fall into three groups:

- *Factual celebrity news.* Headlines like "Jackie Appiah Celebrates Son Damien Agyemang as He Turns 20, Shares Lovely Photos" or "#3MusicAwards21: See The Full List Of Winners" tell you what happened, so I labelled them Informative. But they contain words like *photos*, *video* and celebrity names, which the baseline learned to associate with clickbait (Section 5.2). The model is reacting to the topic and the outlet's style, not to a missing piece of information.
- *Short, dramatic crime headlines.* GhPage headlines like "Wife stabs younger husband to death" or "Police shoots and disarm juju man at Agona Amenfi" are disturbing but state exactly what happened. My codebook says disturbing subject matter is not a clickbait signal by itself (Section 3.3), but the baseline has no way of separating "upsetting" from "withholding".

- *"(Watch Video)" tags.* ZionFelix adds "(Watch Video)" to many headlines that are otherwise complete, for example "Akuapem Poloo In More Trouble As Akuapem Chiefs Disown Her Following Jail Term (Watch Video)". I treated the tag as a call-to-action rather than a curiosity gap, but the model treats it as strong clickbait evidence. Honestly, some of these are borderline, and a different annotator could reasonably have labelled them Clickbait.

Missed clickbait (16 headlines, Clickbait predicted as Informative). All 16 came from the gossip sites, and most of them are exactly the subtle cases my codebook revisions were about:

- *Vague hype with nothing checkable* (revision 3): "Rashida Black Beauty Flaunts Her Fine Figure in a Top and Bodycon Skirt…", "Tracey Boakye Opens a Plush Restaurant in Trassaco…", "John Dumelo's First Son Looks Big And Tall During His 7th Birthday Photoshoot". These are ordinary-looking sentences with names, so they look informative on the surface.
- *Quotes with no anchoring fact* (revision 2): "Sleeping with another person while in a relationship is not cheating – Okyeame Kwame" and "'I Dumped My Obroni Hubby After 3 Weeks in UK for Ghanaian Lover in Ghana': Socialite Afrah Recounts".
- *Vague teasers:* "More videos from Jacobu cheating incident surface online" and "Headmaster discovered dead inside the guest house" (which headmaster? where?).
- *False urgency the model cannot see:* "BREAKING- Parliament passes anti-LGBTQ bill". The capitalised "BREAKING" is one of the signals in my codebook, but TF-IDF lowercases all text by default, so the model never sees the capitals. This is a concrete design flaw in my baseline. Capitalisation should have been kept as a separate feature.

What this tells me. The baseline's errors split cleanly: it over-flags headlines that *look* like gossip and under-flags clickbait that *reads* like a normal sentence. Both point the same way as the held-out-outlet test (Section 5.4). A bag-of-words model learns the vocabulary of gossip sites, not the curiosity-gap structure that actually defines clickbait. The missed cases are also the ones that took me three codebook revisions to pin down, and the same kinds of headline the AI pre-labeller missed most often (Section 3.3), so they are genuinely hard rather than just model failures. A few of the false alarms are arguably labelling judgement calls, which is consistent with the low inter-annotator agreement in Section 3.4.

7. Discussion

RQ1 and RQ2. A simple TF-IDF + Logistic Regression model can do a reasonable job (macro F1 0.647), but only after correcting for class imbalance. Before that, it looked like the best model in the project by accuracy while being useless in practice. Fine-tuned DistilBERT was better on every metric that matters, and the improvement didn't cost much: under two minutes of free GPU time. For a dataset of only about 800 training headlines, I was surprised how big the difference was.

RQ3. Zero-shot classification is not ready to replace a trained model for this task, at least not with BART-MNLI. Its best result roughly matched the baseline, but that best result depended completely on how I worded the labels, and the wording from my original plan gave a model that never found any clickbait. If I had only run the literal version, I would have reported that zero-shot classification "doesn't work" at all, which isn't quite true either. The lesson I took from this is that a zero-shot result is really a result about the model *plus* the prompt, and anyone reporting one should try more than one wording.

RQ4. The held-out-outlet test was the most interesting part of the project for me. The random-split results alone would have made the baseline look more capable than it is. Once each outlet was held out, it became clear that the baseline leans on topic and outlet vocabulary (politicians' names, specific celebrities), while DistilBERT seems to have picked up something more general. I think any clickbait study that compares outlets should include a test like this, given how often earlier datasets took clickbait and non-clickbait from different sources (Section 2).

What clickbait means. The low κ was disappointing at first, but I think it points to something real. Without a codebook, my second annotator treated a lot of ordinary entertainment and celebrity news as clickbait. With the codebook, I only called a headline clickbait when it was holding back information on purpose. Both are reasonable everyday meanings of the word, so "clickbait" is not a well-defined category until you write down exactly what you mean by it. My own codebook only settled after three rounds of changes, and each change came from a real headline that didn't fit. So this task needs a written, tested definition, and the reliability of the labels depends on annotators being trained on it. The κ I measured should not be read as the reliability of the codebook itself. That would need a second, trained annotator, which I didn't have time for.

The AI pre-labels. Using an AI model to draft labels saved time, but it didn't replace the human pass. The AI under-called clickbait badly, catching only 32 of the 196 headlines I labelled

Clickbait. Most of what it missed were exactly the culturally specific cases in my codebook revisions: decontextualised quotes, emotional teasers about disturbing events, and vague hype about celebrities' houses and cars. A dataset built from the AI's labels alone would have been about 94% Informative and would have taught the models a much narrower idea of clickbait.

8. Limitations

- **Small dataset.** 1,000 headlines, 196 of them clickbait, is small. The test set has only 39 clickbait headlines, so a difference of a few headlines changes the recall noticeably, and I didn't compute confidence intervals or run multiple random seeds.
- **One main annotator.** The ground-truth labels are mine. I followed a written codebook, but my judgement is still built into the dataset, and the only agreement check I have was done with an untrained second annotator ($\kappa = 0.057$).
- **Uneven sources.** Graphic Online contributed only 35 headlines, Myjoyonline's headlines come from the first page of five categories, and Yen's come from one slice of its sitemap. Two well-known gossip sites were excluded because of their `robots.txt` rules. The dataset is a snapshot from one day in August 2026, so it might not reflect how these outlets write at other times.
- **Topic imbalance.** Almost all clickbait is entertainment and celebrity news, and almost all politics is informative, so the models can partly confuse topic with clickbait (as the baseline's features show).
- **Headlines only.** I didn't use the article text. Whether a headline is misleading sometimes depends on whether the article actually delivers what the headline promises, and my labels can't capture that.
- **One zero-shot model and two wordings.** I only tried BART-MNLI with two label wordings. A larger generative model, or more careful prompt testing, might do much better.
- **No saved per-headline predictions for DistilBERT and zero-shot.** The Kaggle runs saved metrics and confusion matrices but not each headline's prediction, so my error analysis (Section 6) is done on the baseline. The DistilBERT confusion matrix suggests a similar pattern, but I can't confirm that headline by headline.
- **English only.** Ghanaian headlines sometimes mix in Twi or Pidgin words. I didn't handle these in any special way.

9. Ethics

I only scraped publicly available pages, respected each site's `robots.txt` (including dropping two sites that block AI crawlers), waited 1.5 seconds between requests, and collected only headlines and URLs. I didn't collect any personal data about readers or authors. The headlines do mention public figures, but only as they already appear in public news.

The second annotator received only anonymous IDs and headline text. The file linking those IDs to URLs and outlets stayed with me. They are not named in this report.

Labelling a headline "clickbait" is a judgement about writing style. It is not a claim that the story is false, and the outlet-level numbers in this report should not be read as a ranking of which outlets are trustworthy. A tool like this should only be used to warn readers, not to block or censor content, especially given the false-alarm rate in Section 5.1.

Use of AI tools. As described in Section 3.3, I used an AI model (Claude Code) to produce draft labels, which I reviewed and corrected myself. I also used it as a coding assistant while writing the scraper, the annotation scripts and the notebooks, and to help draft this paper. I verified its output in three ways: every one of the 1,000 AI draft labels was checked by me by hand (and 16.7% were overridden); the code is covered by unit tests and every notebook was re-run end to end with its outputs saved in the repository; and every number in this paper was checked against the saved results files listed in Appendix A. The research design, the labelling decisions, the codebook changes and the interpretation of the results are my own. The full disclosure is in `docs/AI_USE_DISCLOSURE.md` in the repository.

10. Conclusion

I built a labelled dataset of 1,000 Ghanaian news headlines and used it to compare three ways of detecting clickbait. Fine-tuned DistilBERT was clearly the best, with a macro F1 of 0.714, and unlike the TF-IDF baseline it kept most of its performance on outlets it had never seen. Zero-shot classification with BART-MNLI could match the baseline, but only with carefully worded labels, and failed completely with the obvious ones. Along the way I learned that building the dataset was the hardest and most important part. My labels changed a lot once I had to defend them, and someone labelling without the codebook disagreed with me most of the time.

If I continued this project, I would (1) train a second annotator on the codebook and measure agreement properly, (2) collect more headlines, especially clickbait from mainstream outlets

and informative headlines from gossip sites, to reduce the topic shortcut, (3) try multilingual models that can handle Twi and Pidgin, and (4) test larger generative models in the zero-shot setting.

References

Chakraborty, A., Paranjape, B., Kakarla, S., & Ganguly, N. (2016). Stop clickbait: Detecting and preventing clickbaits in online news media. In *2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)* (pp. 9–16). IEEE.

Chen, Y., Conroy, N. J., & Rubin, V. L. (2015). Misleading online content: Recognizing clickbait as "false news". In *Proceedings of the 2015 ACM Workshop on Multimodal Deception Detection* (pp. 15–19). ACM.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement, 20*(1), 37–46.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186). Association for Computational Linguistics.

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences, 120*(30), e2305016120.

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics, 33*(1), 159–174.

Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V., & Zettlemoyer, L. (2020). BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 7871–7880).

Loewenstein, G. (1994). The psychology of curiosity: A review and reinterpretation. *Psychological Bulletin, 116*(1), 75–98.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

Potthast, M., Gollub, T., Hagen, M., & Stein, B. (2018). The Clickbait Challenge 2017: Towards a regression model for clickbait strength. *arXiv preprint arXiv:1812.10847*.

Potthast, M., Köpsel, S., Stein, B., & Hagen, M. (2016). Clickbait detection. In *Advances in Information Retrieval: 38th European Conference on IR Research (ECIR 2016)* (pp. 810–817). Springer.

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

Williams, A., Nangia, N., & Bowman, S. R. (2018). A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of NAACL-HLT 2018* (pp. 1112–1122).

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., et al. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP 2020: System Demonstrations* (pp. 38–45).

Yin, W., Hay, J., & Roth, D. (2019). Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach. In *Proceedings of EMNLP-IJCNLP 2019* (pp. 3914–3923).

Appendix A: GitHub Evidence

Repository: **https://github.com/Mackey-Kumi/ghana-clickbait-detection** (branch `main`). Every number in this report can be traced to a file in the repository:

What	Where in the repo	Commit
Scraper code and unit tests	`scraper/`, `tests/`	`e3efe70` to `5c2d7c1`
Raw scraped headlines (Table 1)	`data/raw/headlines_combined.csv`, `data/raw/<outlet>.csv`, `data/raw/DATA_CARD.md`	`674188a`
Annotation codebook (and its three revisions)	`docs/ANNOTATION_CODEBOOK.md`	`f938fc5`, `00798b9`
AI draft labels with reasons	`data/processed/pre_labels.csv`	`442ffc1`
My verified labels, overrides (Table 2)	`data/processed/verified_labels.csv`	`00798b9`
Second annotator's completed labels	`data/processed/second_annotator_subset.xlsx`	`00798b9`
Cohen's κ computation	`annotate/kappa.py` (run: `python -m annotate.kappa data/processed/second_annotator_subset.xlsx`)	`ac316c7`, `00798b9`

What	Where in the repo	Commit
Final dataset (Table 3)	`data/processed/annotated_headlines.csv`	`00798b9`
Shared train/test split (seed 42)	`models/data.py`, `tests/test_data.py`	`63e371a`
Baseline results, features, held-out test (Tables 4–5, Figure 1)	`notebooks/baseline.ipynb`	`cb54533`
DistilBERT training script and Kaggle logs	`kaggle/experiments/01_finetune_distilbert/`	`9eda7be`
DistilBERT results (Tables 4–5, Figure 2)	`notebooks/transformer.ipynb`, `kaggle/experiments/01_finetune_distilbert/output/`	`9eda7be`
Zero-shot script, logs and results (Table 4, Figure 3)	`kaggle/experiments/02_zeroshot_classification/`, `notebooks/zeroshot.ipynb`	`201d114`
Error analysis (Section 6)	`notebooks/error_analysis.ipynb`	final submission commit
Literature matrix	`docs/LITERATURE_MATRIX.md`	final submission commit
Model card	`docs/MODEL_CARD.md`	final submission commit
Ethics statement and AI-use disclosure	`docs/ETHICS.md`, `docs/AI_USE_DISCLOSURE.md`	final submission commit
Failure log and reproducibility checklist	`docs/FAILURE_LOG.md`, `docs/REPRODUCIBILITY.md`	final submission commit

The submitted version is tagged `v1.0-submission` in the repository.